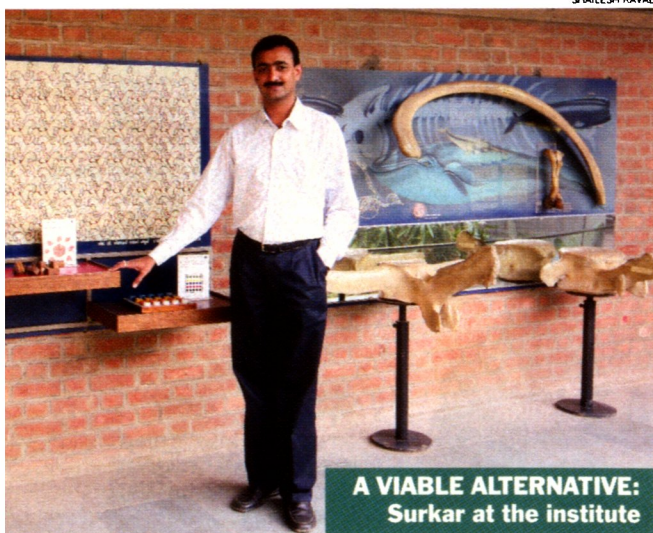


VASCSC, AHMEDABAD

PLAY SCHOOL

If young children today can learn mathematics in a laboratory and science on the playground, then the credit goes to the Vikram Sarabhai Community Science Centre (VASCSC), Ahmedabad. Concerned with the poor quality of science education, eminent scientist Vikram Sarabhai enlisted professors of the Physical Research Laboratory, Ahmedabad, and initiated a movement to cultivate a scientific temper among children. The result was VASCSC in 1966 where children learn science at community centres. Its biggest contribution is the open laboratory concept—dealing with physics, chemistry, biology, electronics, mathematics, model rocketry and



SHAIKESH RAVAL

A VIABLE ALTERNATIVE:
Surkar at the institute

"Teachers from the UK and US visit us to learn techniques of teaching science." DILIP SURKAR, DIRECTOR, VASCSC

computers—where any child can walk in. Nearly 40,000 children use this facility every year. "VASCSC is a laboratory for evolving new ways to teach science to children," says Director Dilip Surkar. The Centre even asked it to manage the German exhibition, the Science Tunnel in India. But the Central and the state government have stopped allocating funds, forcing it to rely on own resources. As they say, it happens only in India.

by Uday Mahurkar



MAGIC MILK: A worker at an automated model milk plant

REUBEN SINGH

NATIONAL DAIRY RESEARCH INSTITUTE, KARNAL

URBAN COWBOYS

This is probably the only institute that trained Mahatma Gandhi. In the summer of 1927, he spent two weeks "as a farmer" at the Imperial Institute for Animal Husbandry and Dairying, Bangalore—a picture of the Mahatma posing with Jill, a crossbred cow, is one of the institute's prized possessions. Rechristened as the National Dairy Research Institute (NDRI) and relocated to Karnal, Haryana, in 1955, it made India the world's highest milk producer and turned dairying into a catalyst for the uplift of the rural masses. By cross-breeding indigenous cows, Sahiwal and Tharparkar, with the best European breeds Holstein-Friesian and Jersey respectively, producing new strains, Karan-Fries and Karan Swiss, with peak milk yields of 46.5 and 44 litre per day, India's milk production rose from 17 million tonne in 1952 to 100 million tonne today. Tailored genetically for Indian agro-climatic conditions, these black-and-white breeds found ready acceptance with farmers. NDRI has also developed cutting-edge technologies for milk processing and human resource for the dairy industry, which at 5 per cent a year is growing faster than agriculture. The 560-hectare campus has an elite herd of over 1,600 cows, buffaloes and goats. The NDRI also has the distinction of producing the world's first test-tube buffalo calf, Pratham. Equipped with state-of-the-art laboratories, the institute has developed a technique to produce more than 10 calves from an elite female cow in one calendar—a method that is, "poised to pave the way for another major breakthrough in milk productivity," according to NDRI Director A.K. Srivastava.

by Ramesh Vinayak

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million cattle and buffalo population in India is the largest in the world, yet the country accounts for less than 1 per cent of the international milk trade.